



CIE A Level Chemistry



Your notes

7.2 Brønsted–Lowry Theory of Acids & Bases

Contents

- * Acids & Bases
- * Brønsted–Lowry Theory
- * Acid & Base Dissociation
- * pH Scale
- * Strength of Acids & Bases
- * pH Titration Curves
- * Indicators used in Titration

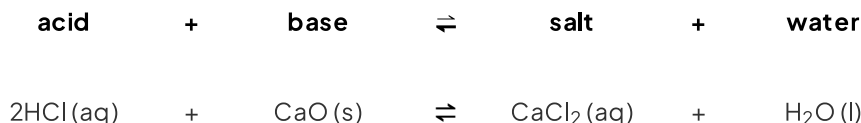


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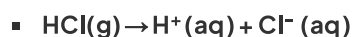
Acids & Bases

Common Acids

- An **acid** is a substance that **neutralises** a base forming a **salt** and **water**:



- Acids are also substances that release **hydrogen ions** when they dissolve in water:



Acid dissociation



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Acids dissociate in water to release a hydrogen ion

Common acids table

Name	Formula	Ions formed in water
Hydrochloric acid	HCl	$\text{H}^+ + \text{Cl}^-$
Nitric acid	HNO_3	$\text{H}^+ + \text{NO}_3^-$
Sulfuric acid	H_2SO_4	$\text{H}^+ + \text{SO}_4^{2-}$
Ethanoic acid	CH_3COOH	$\text{H}^+ + \text{CH}_3\text{COO}^-$

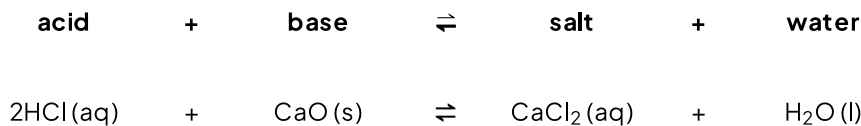
- Monoprotic inorganic acids, such as hydrochloric acid, fully dissociate into their ions
- Organic acids, such as carboxylic acids, do not fully dissociate into their ions
 - Only some of the hydrogen atoms can form ions



Your notes

Common Alkalis

- A **base** is a compound that **neutralises** an acid forming a **salt** and **water**



- A base is a substance that **accepts** hydrogen ions or a compound that contains **oxide** or **hydroxide** ions
 - For example, when the base ammonia is added to water, the ammonium ion and hydroxide ions are formed:

$$\text{NH}_3(\text{g}) + \text{H}_2\text{O (l)} \rightarrow \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$$
 - For example, when sodium hydroxide is dissolved in solution, sodium ions and hydroxide ions are formed:

$$\text{NaOH (s)} + \text{aq} \rightarrow \text{Na}^+(\text{aq}) + \text{OH}^-(\text{aq})$$

- A base that is **soluble** in water is called an **alkali**

Common alkalis table

Name	Formula	Ions formed in water
Sodium hydroxide	NaOH	$\text{Na}^+ + \text{OH}^-$
Potassium hydroxide	KOH	$\text{K}^+ + \text{OH}^-$
Aqueous ammonia	NH_3	$\text{NH}_4^+ + \text{OH}^-$



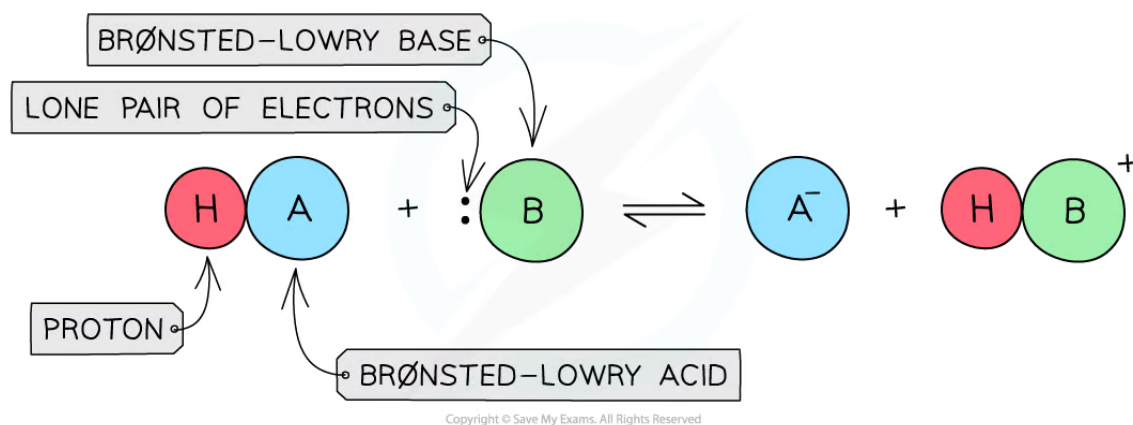
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Brønsted-Lowry Theory

Brønsted-Lowry Theory

- The **Brønsted-Lowry Theory** defines acids and bases in terms of proton transfer between chemical compounds
- A **Brønsted-Lowry acid** is a species that **gives away** a proton (H^+)
- A **Brønsted-Lowry base** is a species that **accepts** a proton (H^+) using its **lone pair of electrons**

How an acid acts as a Brønsted-Lowry proton donor



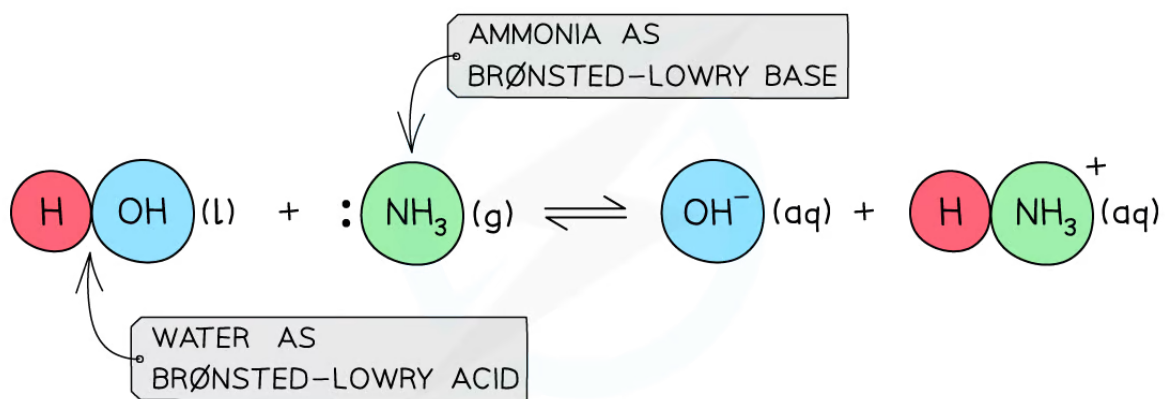
The diagram shows a Brønsted-Lowry acid which donates the proton to the Brønsted-Lowry base that accepts the proton using its lone pair of electrons

- Species that can act both as acids and bases are called **amphoteric**
 - Eg. water as a Brønsted-Lowry acid

Water acting as a Brønsted-Lowry acid



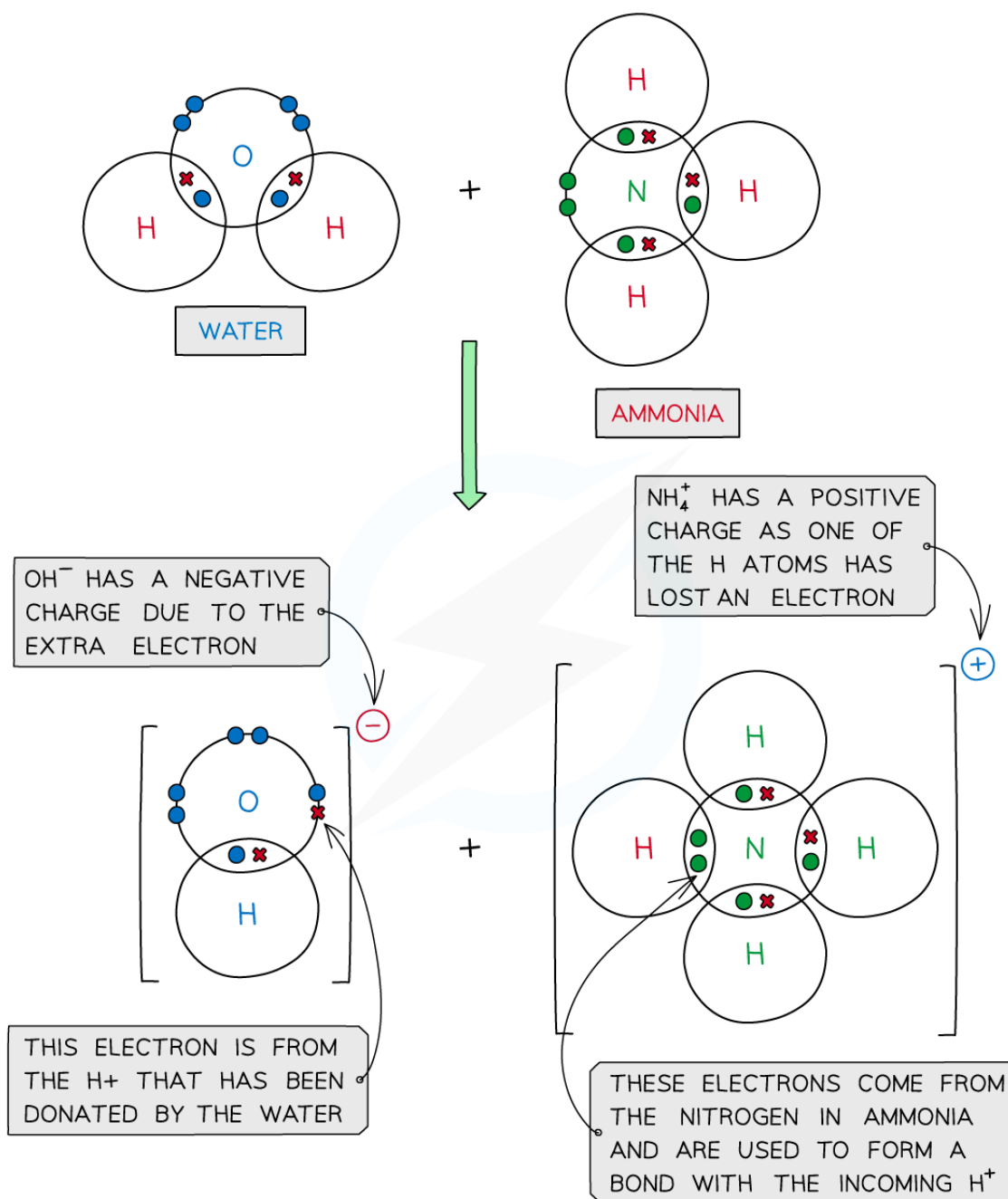
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The diagram shows water acting as a Brønsted-Lowry acid by donating a proton to ammonia which accepts the proton using its lone pair of electrons

Dot and cross diagram showing the Brønsted-Lowry behaviour of water with ammonia

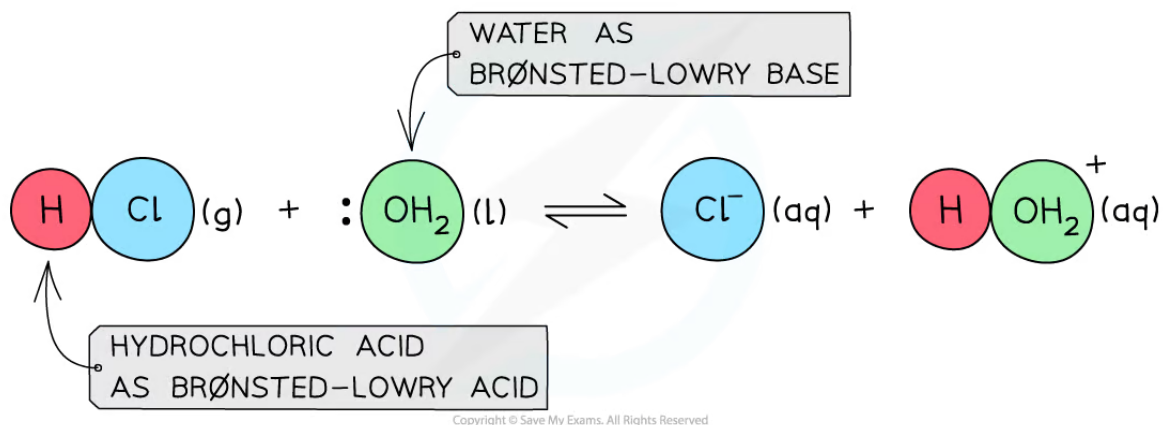


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The diagram shows a dot & cross diagram for the reaction of water with ammonia to show how water acts as a Brønsted-Lowry acid and ammonia as a Brønsted-Lowry base

- Eg. water as a Brønsted-Lowry base

Water acting as a Brønsted-Lowry base

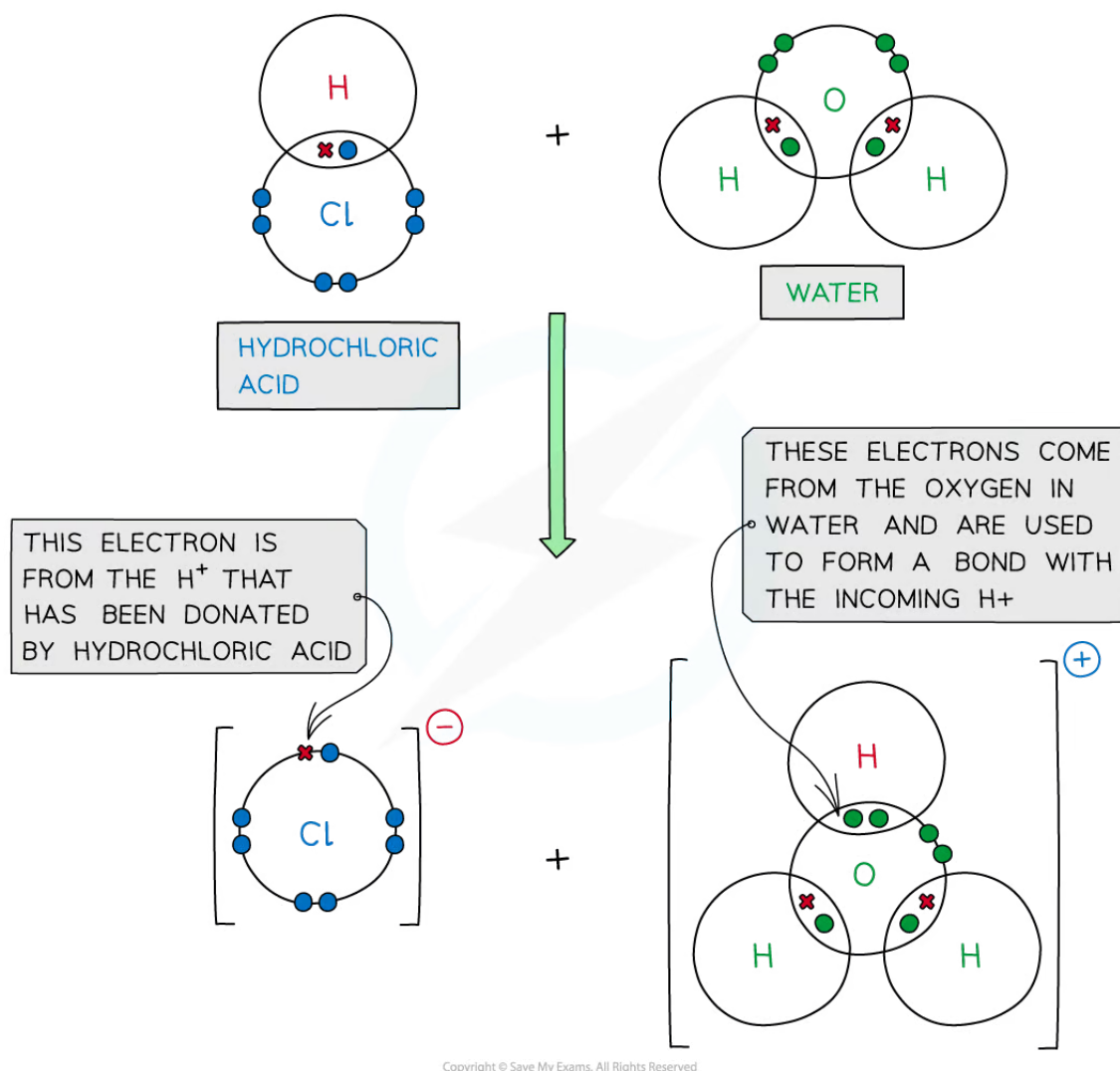


The diagram shows water acting as a Brønsted-Lowry base by accepting a proton from hydrochloric acid proton using its lone pair of electrons

Dot and cross diagram showing the Brønsted-Lowry behaviour of water with hydrochloric acid



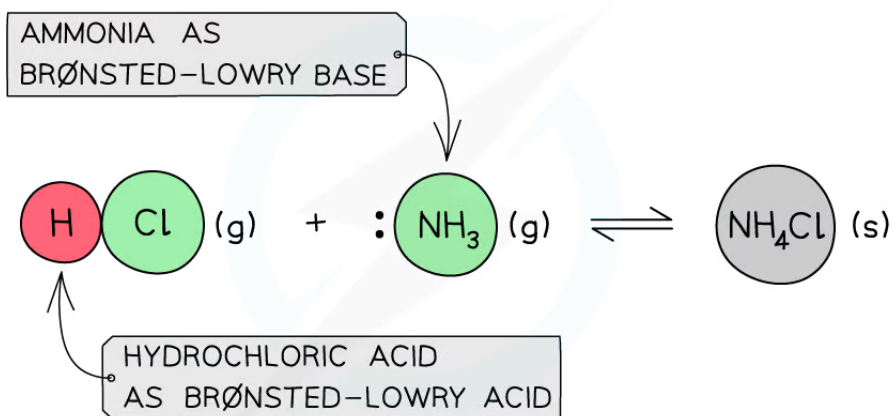
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The diagram shows a dot & cross diagram for the reaction of water with hydrochloric acid to show how water acts as a Brønsted-Lowry base and ammonia as a Brønsted-Lowry acid

- The Brønsted-Lowry Theory is not limited to aqueous solutions only and can also be applied to reactions that occur in the gas phase

A Brønsted-Lowry acid and base reaction



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HCl acts as a Brønsted-Lowry acid by donating a proton while ammonia acts as a Brønsted-Lowry base by accepting a proton

Exam Tip

- An atom of hydrogen contains 1 **proton**, 1 electron and 0 neutrons.
- When hydrogen loses an electron to become **H⁺** only a **proton** remains, which is why a H⁺ ion is also called a proton.



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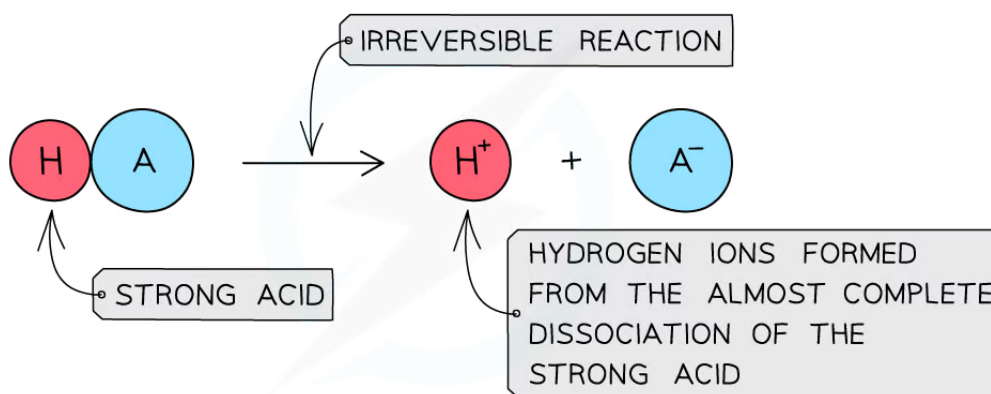
Acid & Base Dissociation

Acid & Base Dissociation

Strong acids

- A **strong acid** is an acid that **dissociates** almost **completely** in aqueous solutions
 - E.g. HCl (hydrochloric acid), HNO₃ (nitric acid) and H₂SO₄ (sulfuric acid)
- The position of the equilibrium is so far over to the **right** that you can represent the reaction as an irreversible reaction

Diagram showing the dissociation of a strong acid in aqueous solution



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In an aqueous solution, a strong acid almost completely dissociates

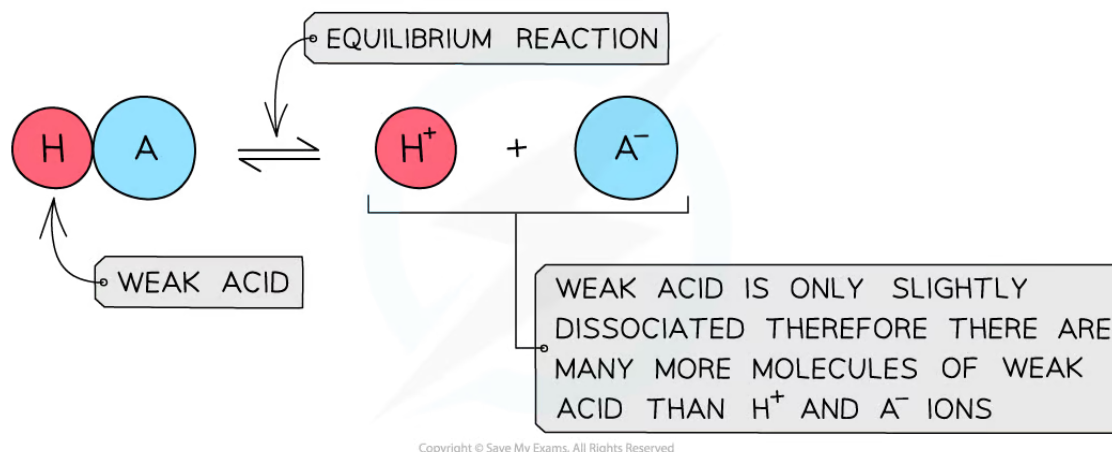
- The solution formed is **highly acidic** due to the high concentration of the H⁺ / H₃O⁺ ions
- Since the **pH** depends on the concentration of H⁺ / H₃O⁺ ions, the pH can be calculated if the concentration of the strong acid is known
 - The concentration of H⁺ / H₃O⁺ ions can be written as [H⁺(aq)]
- pH is the negative log of the concentration of H⁺ / H₃O⁺ ions and can be calculated, if the concentration of the strong acid is known, using the stoichiometry of the reaction

$$\text{pH} = -\log_{10} [\text{H}^+ (\text{aq})]$$

Weak acids

- A **weak acid** is an acid that **partially** (or incompletely) **dissociates** in aqueous solutions
 - E.g. most organic acids (ethanoic acid), HCN (hydrocyanic acid), H₂S (hydrogen sulfide) and H₂CO₃ (carbonic acid)
- The position of the equilibrium is more to the **left** and an equilibrium is established

Diagram showing the dissociation of a weak acid in aqueous solution



In an aqueous solution, a weak acid does not fully dissociate

- The solution is **less acidic** due to the lower concentration of $\text{H}^+ / \text{H}_3\text{O}^+$ ions
- Finding the **pH** of a weak acid is a bit more complicated as now the concentration of H^+ ions is not equal to the concentration of acid
- To find the concentration of H^+ ions, the acid dissociation constant (K_a) should be used

Acid & equilibrium position table

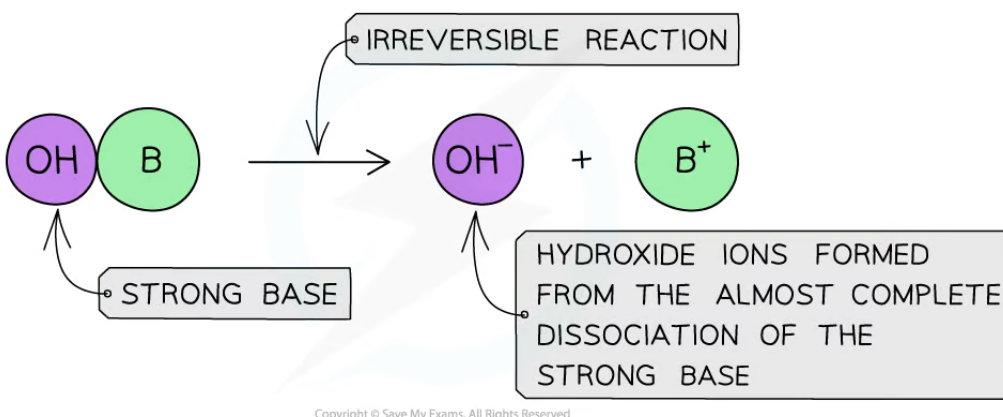
	Strong acid	Weak acid
Position of equilibrium	Right	Left
Dissociation	Completely ($\rightarrow$)	Partially ($\rightleftharpoons$)
H^+ concentration	High	Low
pH	Use [strong acid] for $[\text{H}^+]$	Use K_a to find $[\text{H}^+]$
Examples	HCl HNO_3 H_2SO_4 (first ionisation)	Organic acids, e.g. ethanoic acid HCN H_2S H_2CO_3

Strong bases



- A **strong base** is a base that dissociates almost completely in aqueous solutions
 - E.g. group 1 metal hydroxides such as NaOH (sodium hydroxide)
- The position of the equilibrium is so far over to the right that you can represent the reaction as an irreversible reaction

Diagram showing the dissociation of a strong base in aqueous solution



In an aqueous solution, a strong base almost completely dissociates

- The solution formed is highly basic due to the high concentration of the OH⁻ ions

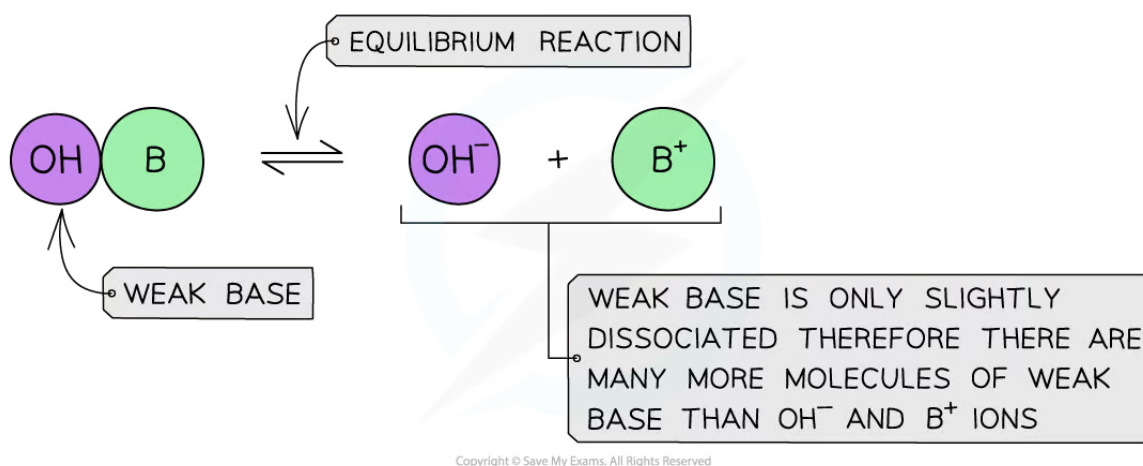
Weak bases

- A **weak base** is a base that **partially** (or incompletely) **dissociates** in aqueous solutions
 - E.g. NH₃ (ammonia), amines and some hydroxides of transition metals
- The position of the equilibrium is more to the **left** and an equilibrium is established

Diagram showing the dissociation of a weak base in aqueous solution



Your notes



In an aqueous solution, a weak base does not fully dissociate

- The solution is **less basic** due to the lower concentration of OH⁻ ions

Base & equilibrium position table

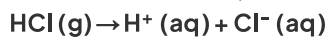
	Strong base	Weak base
Position of equilibrium	Right	Left
Dissociation	Completely (→)	Partially (⇌)
OH ⁻ concentration	High	Low
Examples	Group 1 metal hydroxides	NH ₃ Amines Some transition metal hydroxides



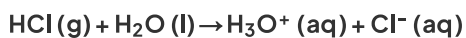
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 Exam Tip

- Hydrogen ions in aqueous solutions can be written as either as H_3O^+ or as H^+ however, if H_3O^+ is used, H_2O should be included in the chemical equation:



or



- Remember that some acids are both strong and weak acids – for example, H_2SO_4 (sulfuric acid) has two hydrogen ions that can ionise.
 - H_2SO_4 acts as a strong acid: $\text{H}_2\text{SO}_4 \rightarrow \text{H}^+ + \text{SO}_4^-$
 - HSO_4^- acts as a weak acid: $\text{HSO}_4^- \rightleftharpoons \text{H}^+ + \text{SO}_4^{2-}$
- Also, don't forget that the terms **strong** and **weak** acids and bases are related to the **degree of dissociation** and not the **concentration**.
 - The appropriate terms to use when describing **concentration** are **dilute** and **concentrated**.



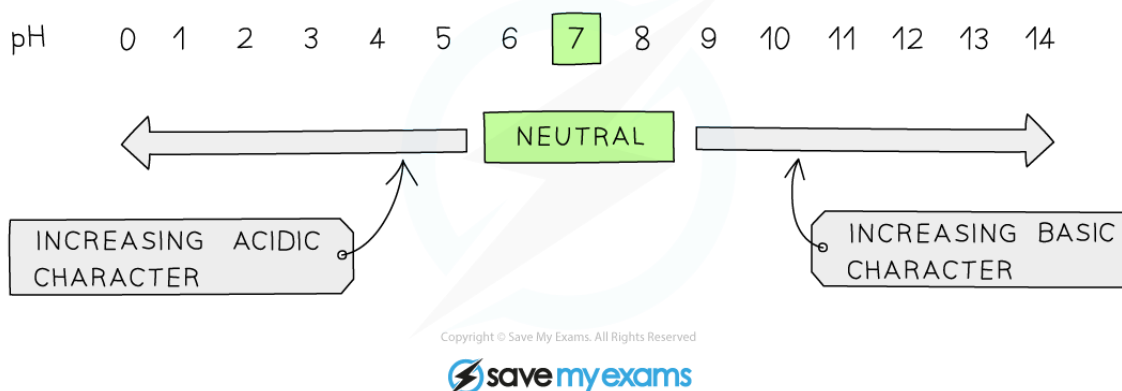
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pH Scale

The pH Scale

- The pH scale is a numerical scale that shows how **acidic** or **alkaline** a solution is
- The values on the pH scale go from 1–14 (extremely acidic substances have values of below 1)
 - All acids have pH values **below** 7
 - All alkalis have pH values **above** 7
- The **lower** the pH, the **more acidic** the solution is
- The **higher** the pH, the **more alkaline** the solution is

The pH scale



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The pH scale shows the acidity, neutrality and alkalinity of chemicals

pH of water

- An equilibrium exists in water where few water molecules dissociate into proton and hydroxide ions:



- The equilibrium constant expression for this reaction is:

$$K_c = \frac{[\text{H}^+] [\text{OH}^-]}{[\text{H}_2\text{O}]}$$

- The equilibrium constant expression can be rearranged to:

$$K_c \times [\text{H}_2\text{O}] = [\text{H}^+] [\text{OH}^-]$$



Your notes

- Since the concentration of the H^+ and OH^- ions is very small, the concentration of water is considered to be a constant

- So, the expression can be rewritten as:

$$K_w = [\text{H}^+] [\text{OH}^-]$$

- Where K_w (ionic product of water) = $K_c \times [\text{H}_2\text{O}] = 10^{-14} \text{ mol}^2 \text{ dm}^{-3}$ at 298K
- Water at 298K has **equal amounts** of OH^- and H^+ ions with concentrations of $10^{-7} \text{ mol dm}^{-3}$
- To calculate the pH of water, the following formula should be used:

$$\text{pH} = -\log [\text{H}^+ (\text{aq})]$$

- Where $[\text{H}^+ (\text{aq})]$ is the concentration of $\text{H}^+ / \text{H}_3\text{O}^+$ ions
- So, the calculation is:
 - $\text{pH} = -\log (10^{-7}) = 7$
- Thus, water has a pH of 7

pH of acids

- Acidic** solutions (strong or weak) **always** have more H^+ than OH^- ions
- Since the concentration of H^+ is always **greater** than the concentration of OH^- ions, $[\text{H}^+]$ is always **greater** than $10^{-7} \text{ mol dm}^{-3}$
- Using the pH formula, this means that the **pH of acidic solutions** is always **below 7**
- The higher the $[\text{H}^+]$ of the acid, the lower the pH

pH of bases

- Basic** solutions (strong or weak) **always** have more OH^- than H^+ ions
- Since the concentration of OH^- is always **greater** than the concentration of H^+ ions, $[\text{H}^+]$ is always **smaller** than $10^{-7} \text{ mol dm}^{-3}$
- Using the pH formula, this means that the **pH of basic solutions** is always **above 7**
- The higher the $[\text{OH}^-]$ of the base, the higher the pH



Your notes

Strength of Acids & Bases

Strong & Weak Acids & Bases

- Strong and weak acids can be distinguished from each other by their:
 - pH value** (using a pH meter or universal indicator)
 - Electrical conductivity**
 - Reactivity**

pH

- An acid **dissociates** into H^+ in solution according to:

$$HA \rightarrow H^+ + A^-$$
- The **stronger** the acid, the **greater** the **concentration of H^+** and therefore the **lower the pH**

pH value of a strong & weak acid table

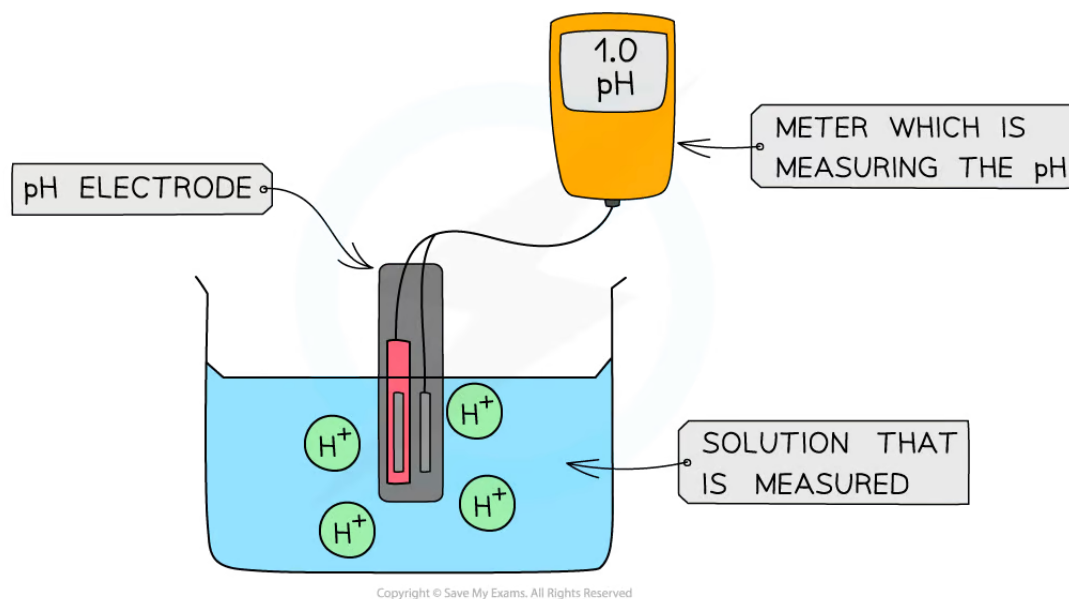
Acid	pH of 0.1 mol dm^{-3} solution
HCl (strong)	1
CH_3COOH (weak)	2.9

- The most **accurate** way to determine the pH is by reading it off a **pH meter**
- The pH meter is connected to the **pH electrode** which shows the pH value of the solution

Using a digital pH meter



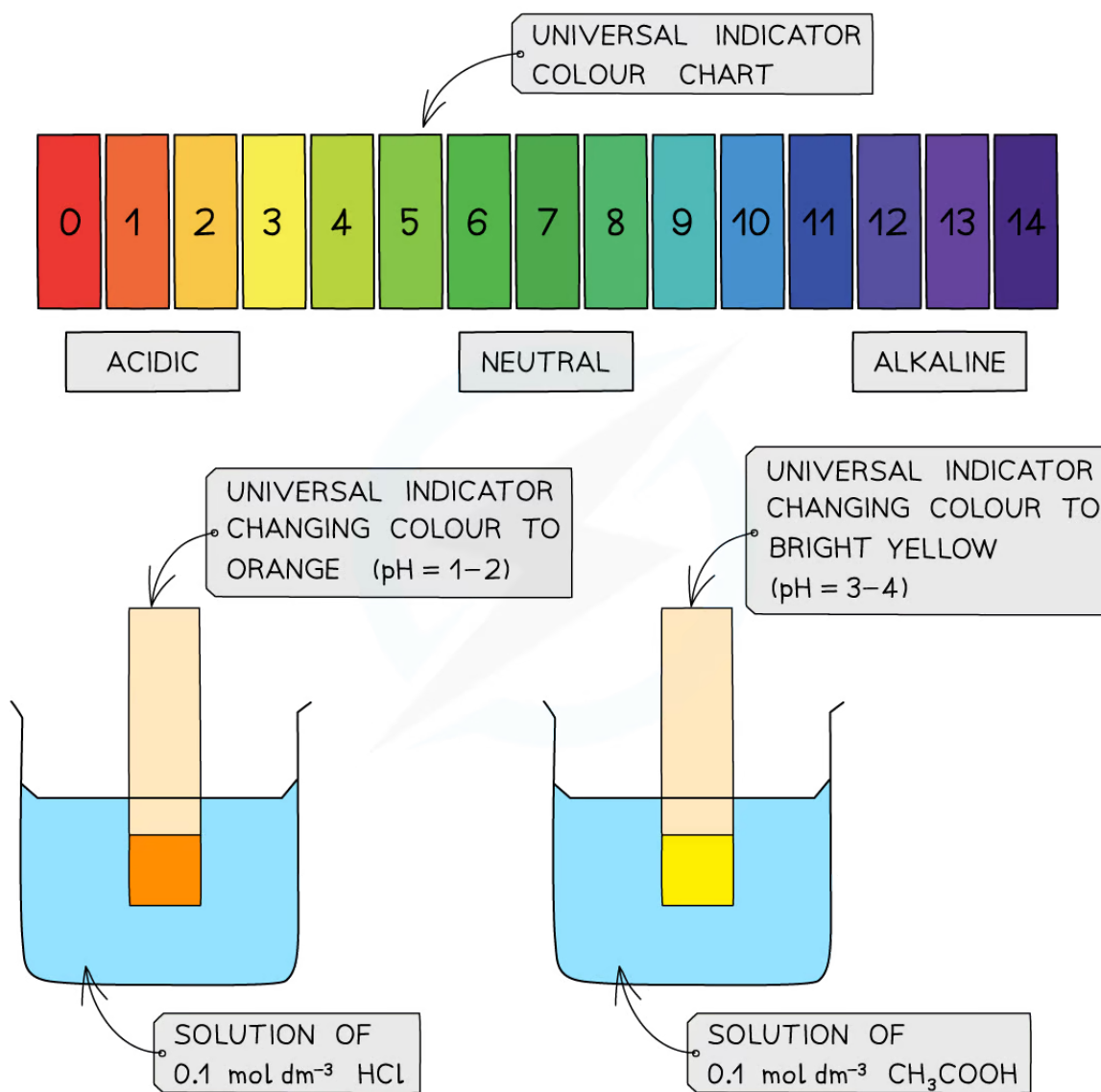
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The diagram shows a digital pH meter measures the pH of a solution using a pH electrode

- A less accurate method is to measure the pH using universal indicator paper
- The universal indicator paper is dipped into a solution of acid upon which the paper changes colour
- The colour is then compared to those on a chart which shows the colours corresponding to different pH values

How to use universal indicator paper



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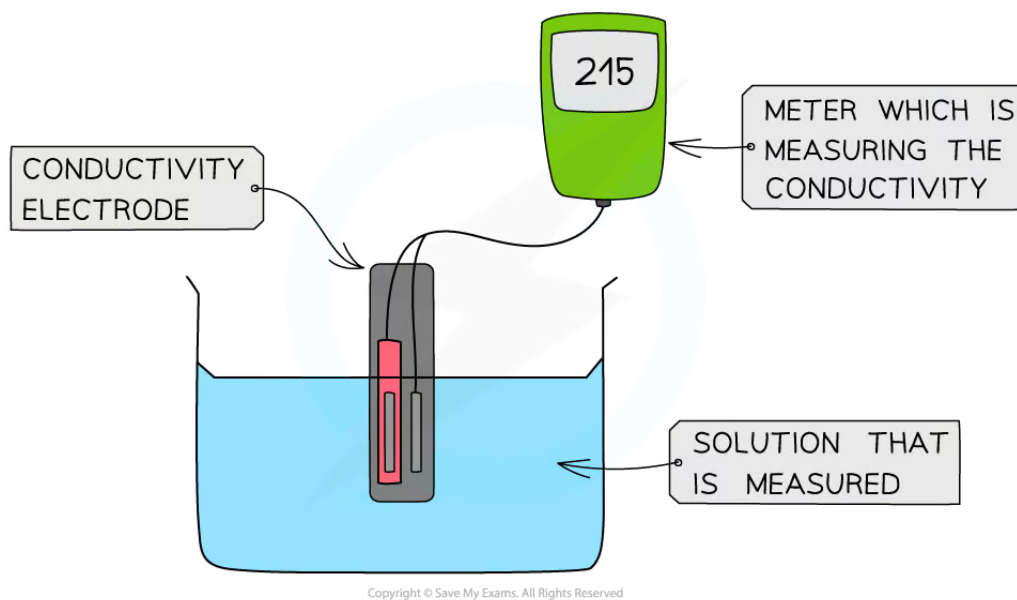
The diagram shows the change in colour of the universal indicator paper when dipped in a strong (HCl) and weak (CH₃COOH) acid. The colour chart is used to read off the corresponding pH values which are between 1–2 for HCl and 3–4 for CH₃COOH

Electrical conductivity

- Since a **stronger acid** has a **higher concentration of H⁺** it **conducts electricity** better
- Stronger acids therefore have a greater **electrical conductivity**

- The electrical conductivity can be determined by using a **conductivity meter**
- Like the pH meter, the conductivity meter is connected to an electrode
- The conductivity of the solution can be read off the meter

Using a digital conductivity meter



The diagram shows a digital conductivity meter that measures the electrical conductivity of a solution using an electrode

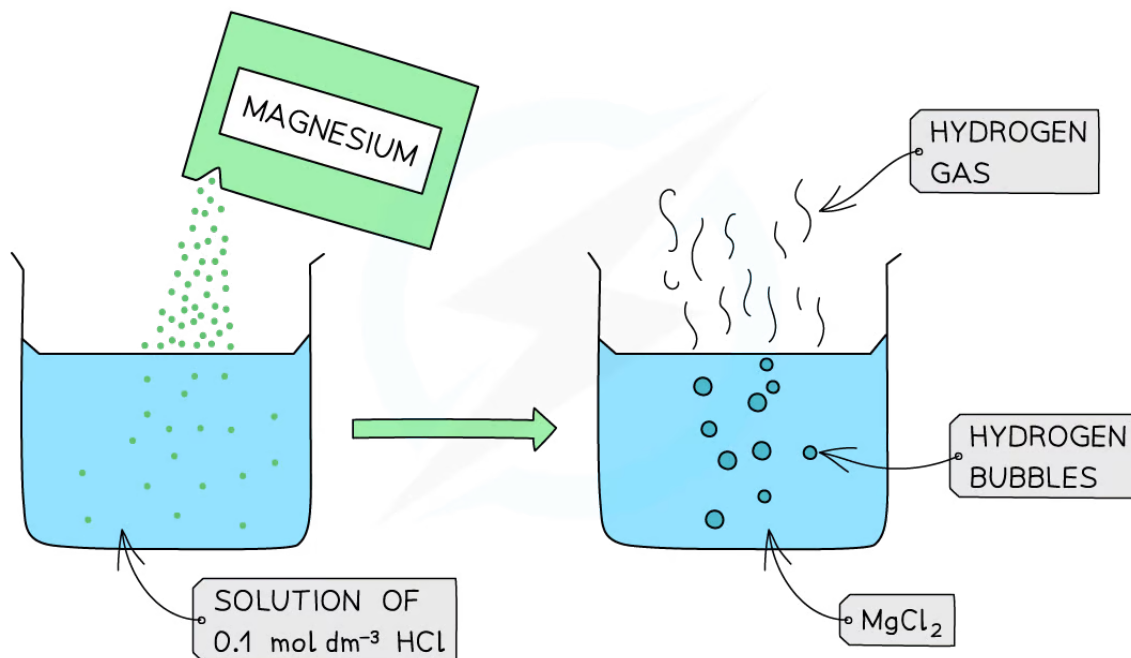
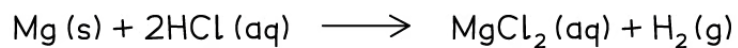
Reactivity

- Strong and weak acids of the same concentrations react differently with reactive metals
- This is because the concentration of H^+ is greater in strong acids compared to weak acids
- The greater H^+ concentration means that more H_2 gas is produced

The reaction of 0.1 mol dm^{-3} of a strong acid, HCl , with Mg



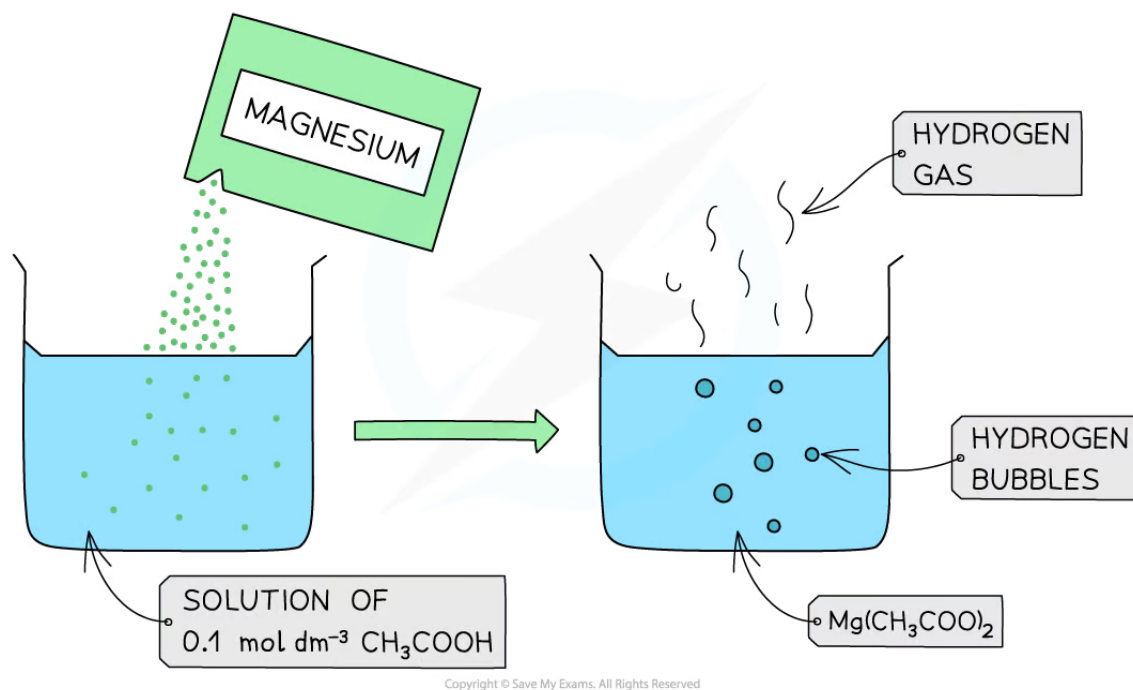
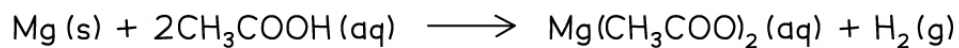
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The reaction produces a lot of bubbles and hydrogen gas due to the high concentration of H^+ present in the solution

The reaction of 0.1 mol dm^{-3} of a weak acid, CH_3COOH , with Mg



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The reaction produces fewer bubbles and hydrogen gas due to the lower concentration of H⁺ present in the solution

Exam Tip

- The above-mentioned properties of strong and weak acids depend on their ability to dissociate and form H⁺ ions.
- Stronger acids dissociate more, producing a greater concentration of H⁺ ions and therefore showing lower pH values, greater electrical conductivity and more vigorous reactions with reactive metals.



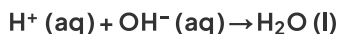
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Neutralisation Reactions

- A neutralisation reaction is one in which an acid ($\text{pH} < 7$) and a base/alkali ($\text{pH} > 7$) react together to form water ($\text{pH} = 7$) and a salt:



- The proton of the acid reacts with the hydroxide of the base to form water:



- The spectator ions which are not involved in the formation of water are $\text{Na}^+ (\text{aq}) + \text{Cl}^- (\text{aq})$

- These react to form the salt:



- The name of the salt produced can be predicted from the acid that has reacted

Table of salts produced from certain acids

Acid reacted	Salt produced
Hydrochloric	Chloride
Sulfuric	Sulfate
Nitric	Nitrate
Ethanoic	Ethanoate

Exam Tip

Note that the reaction of an acid and metal carbonate also forms carbon dioxide:





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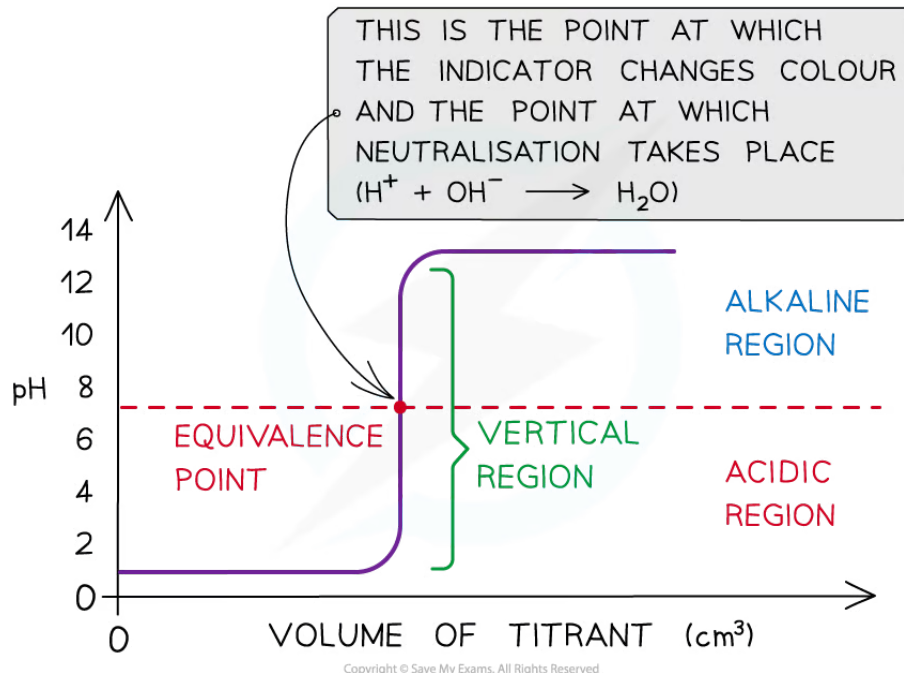
pH Titration Curves

pH Titration Curves

What are pH titration curves?

- Titration is a technique used in neutralisation reactions between acids and alkalis to determine the concentration of the unknown solution
 - It involves adding a **titrant** of known concentration from a burette into a conical flask containing the **analyte** of unknown concentration
 - An indicator is added which will change colour at the **endpoint** of the titration
 - The endpoint is the point at which an equal number of moles of **titrant** and **analyte** react with each other
 - The equivalence point is halfway along the **vertical region** of the curve
- Equivalence point $\rightarrow$ moles of alkali = moles of acid**
- This is also known as the **equivalence point** and this is the point at which **neutralisation** takes place

Example pH titration curve



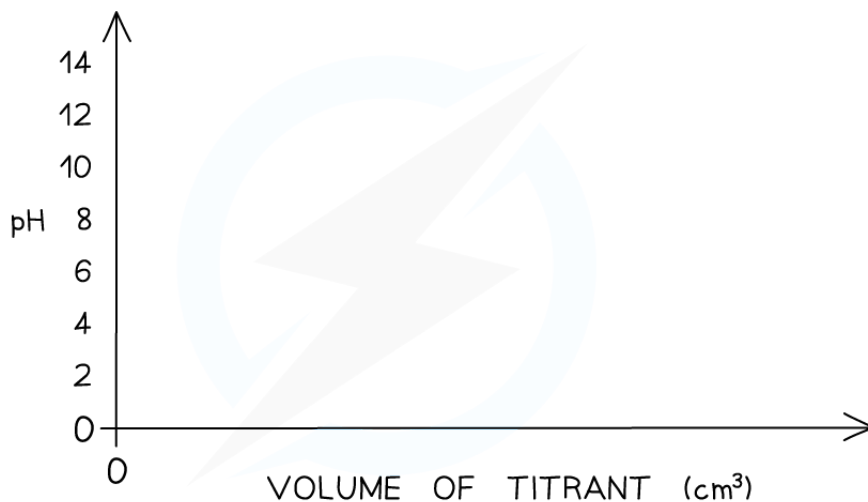
The equivalence point is the point at which an equal number of moles of titrant and analyte have reacted



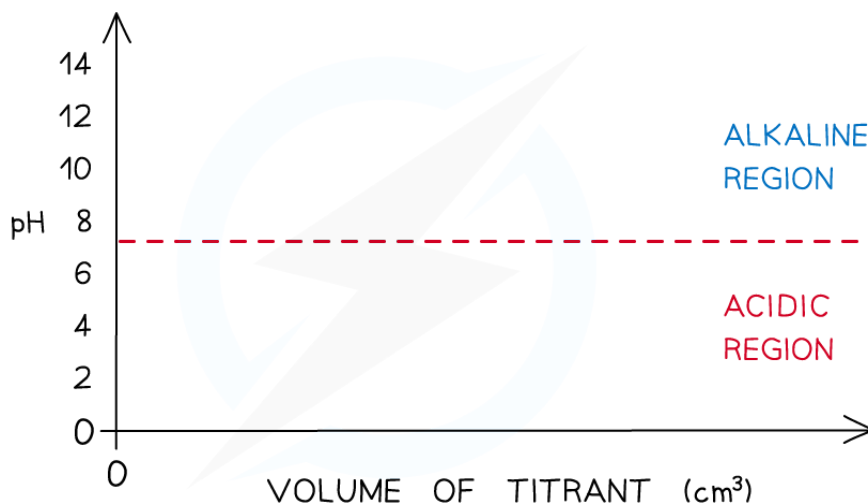
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Sketching a pH titration curve

- Draw axes with volume added (cm^3) on the x-axis and pH on the y-axis



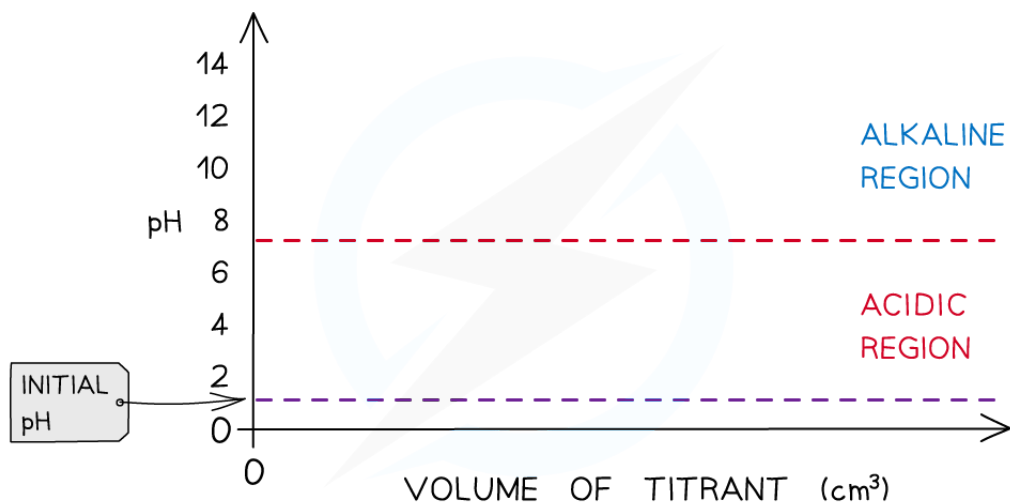
- Draw a horizontal line running parallel to the x-axis at pH 7
 - Everything below this line will be in the acidic **region** and everything above it in the **alkaline** region



- Determine which substance is in the conical flask

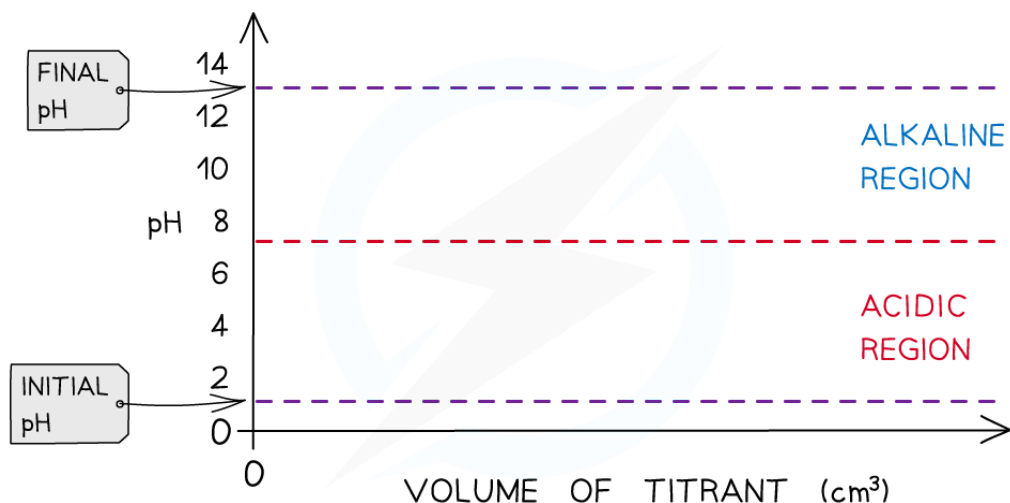
- If it is a **strong acid** the initial pH is about 1 or 2
- If it is a **weak acid** the initial pH is about 2–3
- If it is a **strong alkali** the initial pH is about 13–14
- If it is a **weak alkali** the initial pH is about 11


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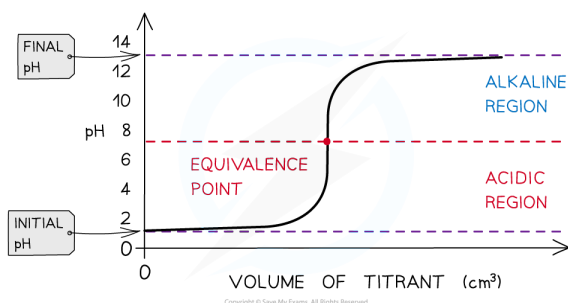
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- Determine what type of acid and alkali are used:
 - Strong acid + strong alkali
 - Strong acid + weak alkali
 - Weak acid + strong alkali
 - Weak acid + weak alkali



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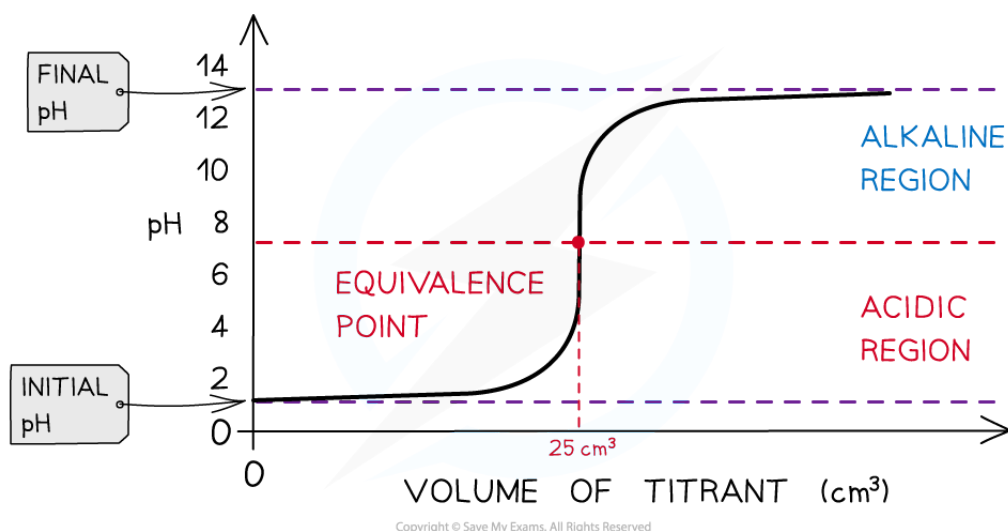
- Draw the pH titration curve



Strong acid + strong alkali pH titration curve

- Initially, there are only H^+ ions present in the solution from the dissociation of the strong acid (HCl) (initial pH about 1–2)
- As the **volume of strong alkali (NaOH) added increases**, the pH of the HCl solution slightly increases too as more and more H^+ ions react with the OH^- from the NaOH to form water
- The change in pH is not that much until the volume added gets close to the equivalence point
- The pH surges upwards very steeply
- The **equivalence point** is the point at which all H^+ ions have been neutralised
 - Therefore, the pH is 7 at the equivalence point
- Adding more NaOH will increase the pH as now there is an **excess in OH^- ions** (final pH about 13–14)

pH titration curve for a strong acid + strong alkali



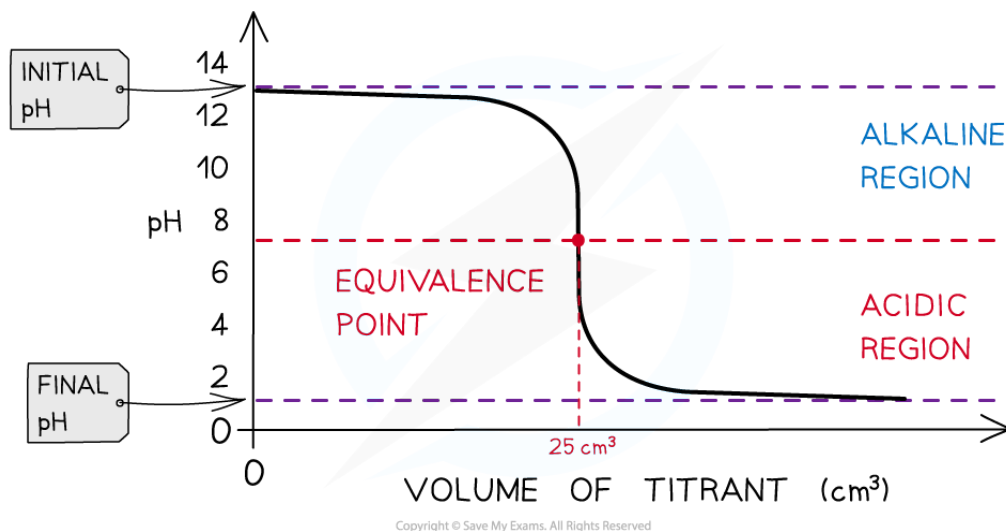


Your notes

The diagram shows a pH titration curve of $1.0 \text{ mol dm}^{-3} \text{ HCl}$ (25 cm^3) with NaOH

- The pH titration curve for HCl added to a NaOH has the **same shape**
- The initial pH and final pH are the other way around
- The equivalence point is still 7

pH titration curve for a strong alkali + strong acid

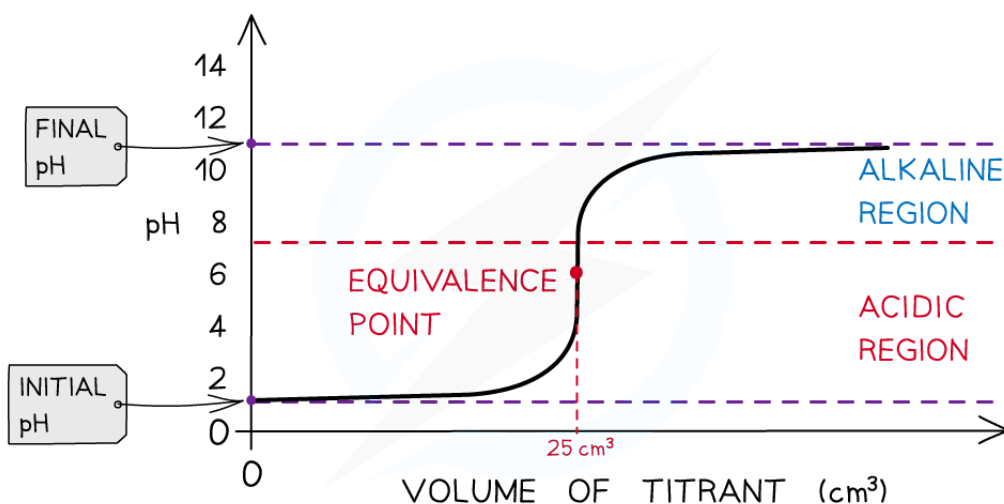


The diagram shows a pH titration curve of $1.0 \text{ mol dm}^{-3} \text{ NaOH}$ (25 cm^3) with HCl

Strong acid + weak alkali pH titration curve

- Initially, there are **only H^+ ions present** in the solution from the dissociation of the strong acid (HCl) (initial pH about 1–2)
- As the volume of weak alkali (NH_3) added increases, the **pH of the analyte solution slightly increases too** as more and more H^+ ions react with the NH_3
- The change in pH is not that much until the volume added gets close to the equivalence point
- The **equivalence point is the point at which all H^+ ions have been neutralised by the NH_3** however the equivalence point is not neutral, but the solution is still acidic (pH about 5.5)
- This is because all H^+ have reacted with NH_3 to form NH_4^+ which is a relatively strong acid, causing the solution to be acidic
- As more of the NH_3 is added, the pH increases to above 7 but below that of a strong alkali as NH_3 is a weak alkali

pH titration curve for a strong acid + weak alkali



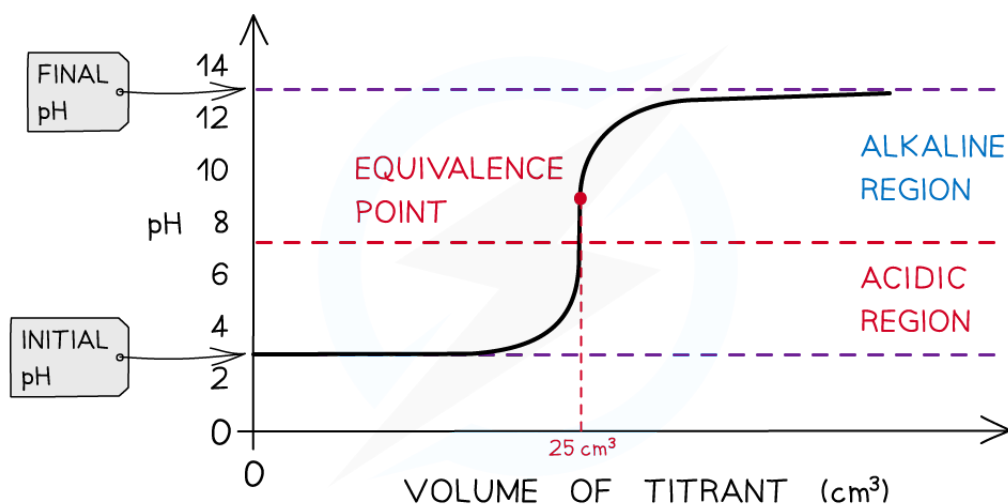
The diagram shows a pH titration curve of 1.0 mol dm⁻³ HCl (25 cm³) with NH₃

- The pH titration curve for strong acid added to a weak alkali has the same shape
- The initial and final pH are the other way around
- The **equivalence point is still about 5.5**

Weak acid + strong alkali pH titration curve

- Initially, there are **only H⁺ ions present in the solution** from the dissociation of the weak acid (CH₃COOH, ethanoic acid) (initial pH about 2–3)
- As the volume of strong alkali (NaOH) added increases, the **pH of the ethanoic acid solution slightly increases too** as more and more H⁺ ions react with the OH⁻ from the NaOH to form water
- The change in pH is not that much until the volume added gets close to the equivalence point
- The pH surges upwards very steeply
- The **equivalence point is the point at which all H⁺ ions have been neutralised by the OH⁻ ions** however the equivalence point is not neutral, but the solution is slightly basic (pH about 9)
- This is because all H⁺ in CH₃COOH have reacted with OH⁻ however, CH₃COO⁻ is a relatively strong base, causing the solution to be basic
- As more of the NaOH is added, the pH increases to about 13–14

pH titration curve for a weak acid + strong alkali



The diagram shows a pH titration curve of a weak acid with a strong base

- The pH titration curve for weak acid added to a strong alkali has the same shape
- The initial and final pH are the other way around
- The **equivalence point** is still about 9

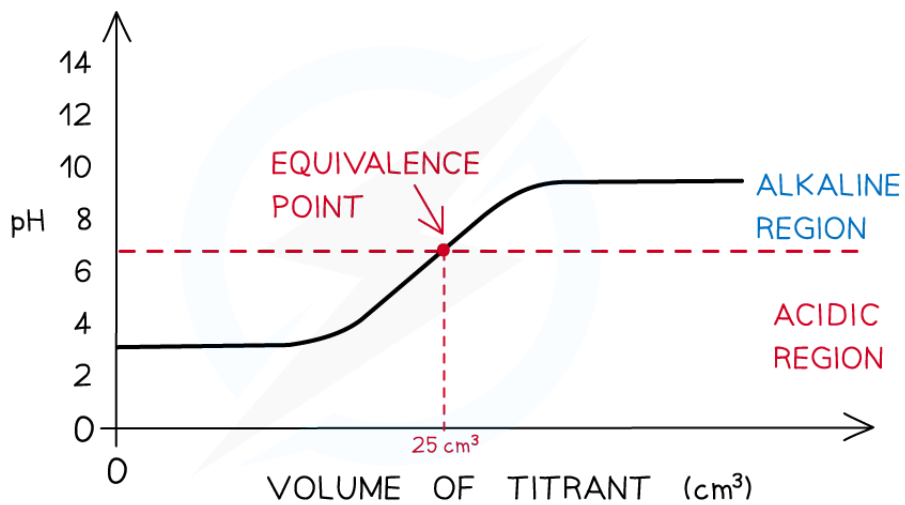
Weak acid + weak alkali pH titration curve

- Initially, there are only H^+ ions present in the solution from the dissociation of the weak acid (CH_3COOH , ethanoic acid) (initial pH about 2–3)
- In these pH titration curves, there is no vertical region
- There is a '**point of inflexion**' at the equivalence point
- The curve does not provide much other information

pH titration curve for a weak acid + weak alkali



Your notes



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The diagram shows a pH titration curve of 1.0 mol dm^{-3} weak acid (25 cm^3) with weak alkali

Exam Tip

You should be able to read and sketch pH titration curves of titrations where the titrant is an acid or an alkali.

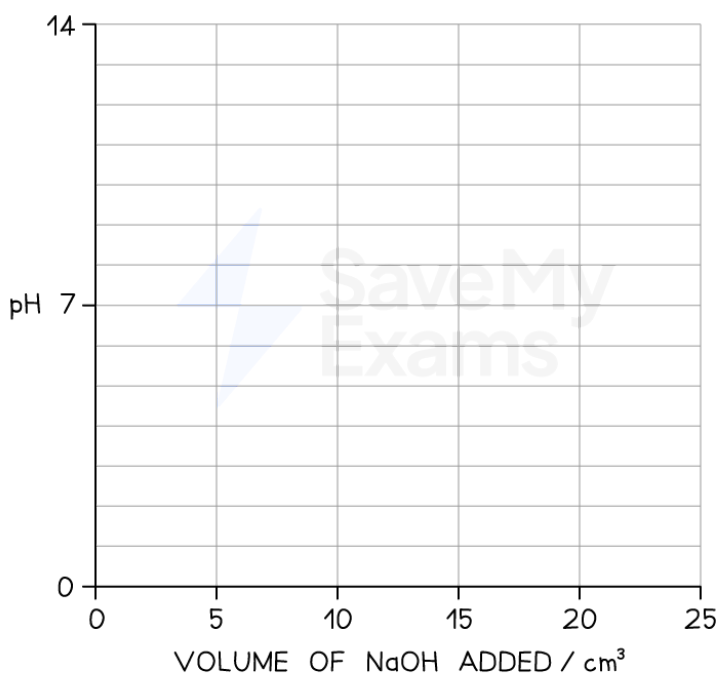


Your notes

Worked example

A 10.0 cm^3 sample of $0.150 \text{ mol dm}^{-3}$ aminoethanoic acid with a pH of 5.3 was titrated with $0.100 \text{ mol dm}^{-3}$ NaOH. After 20.0 cm^3 of NaOH, an excess, had been added, the pH was found to be 12.5.

Using the following axes, sketch a graph showing how the pH changes during this titration.



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[3]

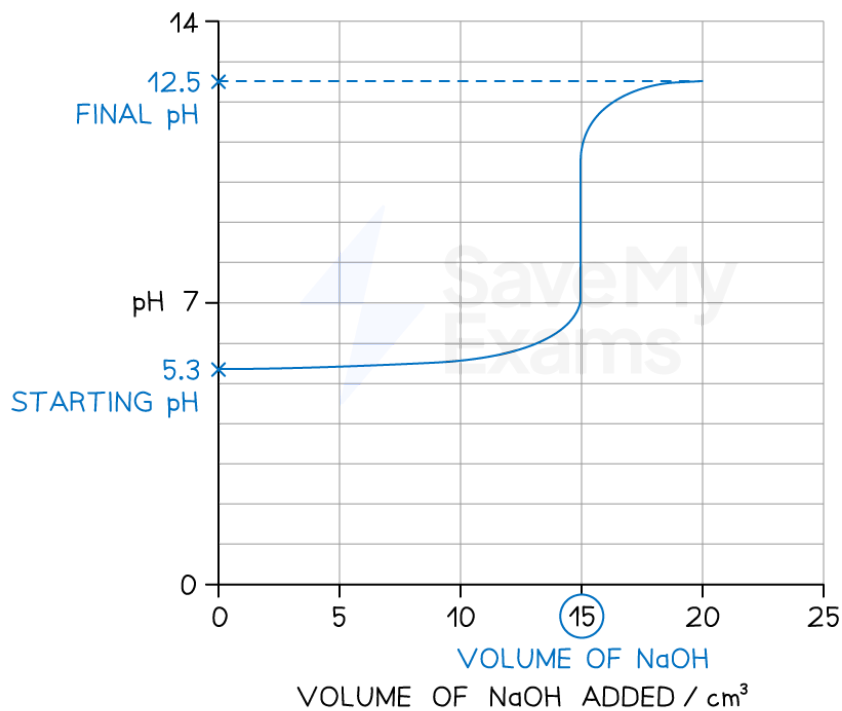
Answer

- The curve starts at pH 5.3
 - Mark on graph
- The volume of NaOH added to reach the vertical section of the graph = 15.0 cm^3
 - Vol acid x Concentration acid = Vol base x Concentration base
 - $10 \times 0.150 = \text{Vol base} \times 0.100$
 - $\frac{(10 \times 0.150)}{0.100} = 15.0 \text{ cm}^3$
 - There is no mark for the height of the vertical section, but the equivalence point must be **above** pH 7 for a weak acid – strong base titration
- The curve finishes at pH = 12.5 at 20 cm^3 .



Your notes

- Make sure the graph does not go above pH 12.5
 - This is the maximum pH value given in the question
- Make sure that the volume does not exceed 20 cm³
 - This is the maximum volume of base added given in the question





Your notes

Indicators used in Titration

Indicators

- **Indicators** are substances that change colour when they are added to acidic or alkaline solutions
- When choosing the appropriate indicator, the pH of the equivalence point is very important
- The two most common indicators that are used in titrations are **methyl orange** and **phenolphthalein**

Indicator & pH range table

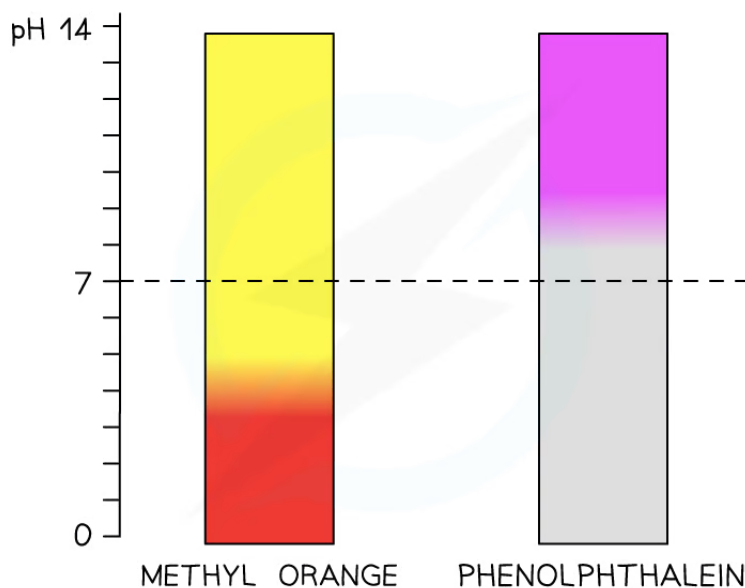
Indicator	pH range
Methyl orange	3.1 – 4.4
Phenolphthalein	8.3 – 10.0

- Both indicators change colour over a specific pH range

Diagram showing the colour changes for methyl orange and phenolphthalein



Your notes



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Methyl orange changes from red to yellow over a pH range of 3.1 – 4.4, while phenolphthalein changes from colourless to pink over a pH range of 8.3 – 10.0

Choosing indicators for titrations

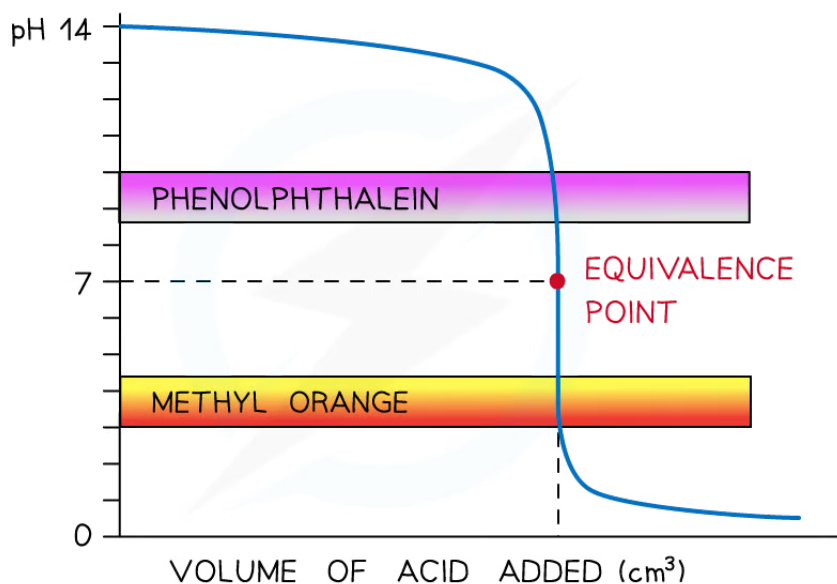
Strong acid and strong alkali

- The colour change for **both indicators** takes place at a pH range that falls within the vertical region of the curve
- Therefore, either indicator can be used

Methyl orange and phenolphthalein in a strong acid + strong alkali titration



Your notes



Both indicators can be used to determine the endpoint of the titration of a strong acid and strong alkali

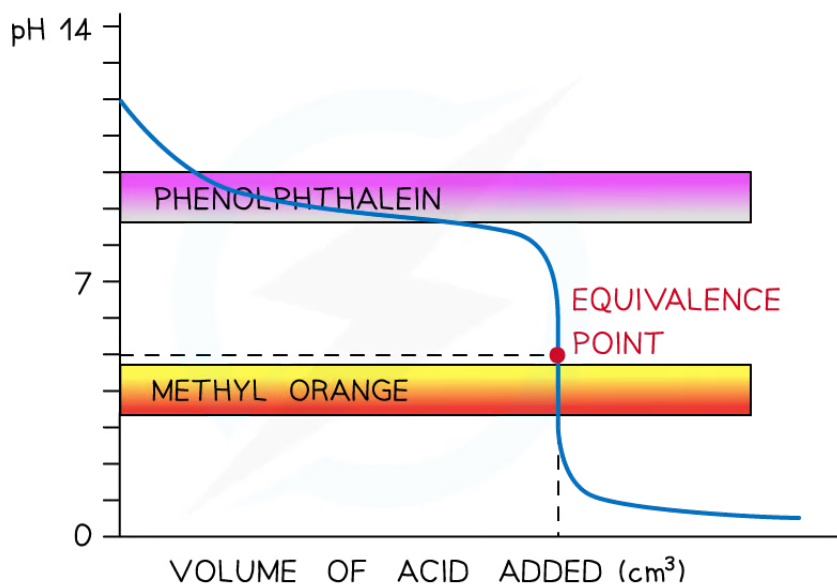
▪ **Strong acid and weak alkali**

- Only **methyl orange** will change colour at a pH close to the equivalence point and within the vertical region of the curve

Methyl orange and phenolphthalein in a strong acid + weak alkali titration



Your notes



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Only methyl orange can be used to determine the endpoint of the titration of a strong acid and weak alkali

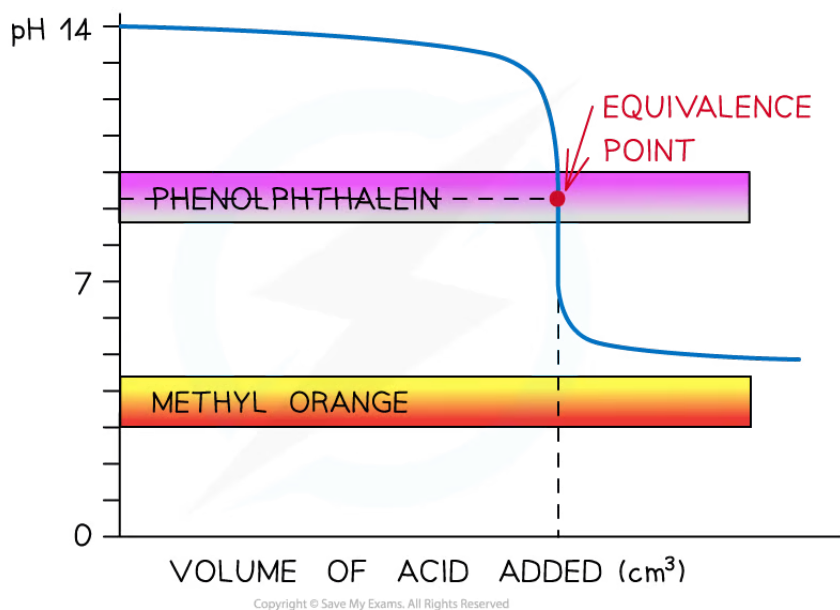
■ **Weak acid and strong alkali**

- Now, only **phenolphthalein** will change colour at a pH close to the equivalence point and within the vertical region of the curve
- The pH range at which methyl orange changes colour falls below the curve

Methyl orange and phenolphthalein in a weak acid + strong alkali titration



Your notes



Only phenolphthalein can be used to determine the endpoint of the titration of a weak acid and strong alkali

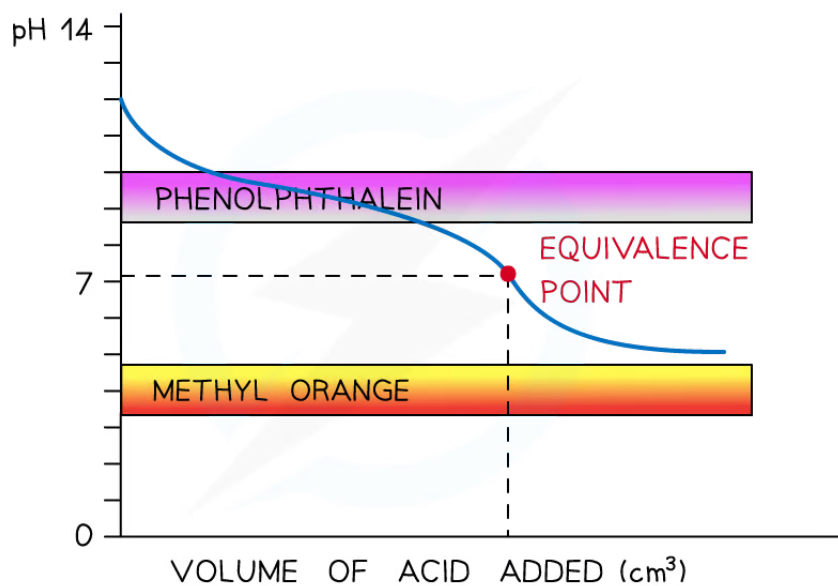
■ **Weak acid and weak alkali**

- Neither indicator is useful, and a different method should be considered

Methyl orange and phenolphthalein in a weak acid + weak alkali titration



Your notes



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Neither indicator can be used to determine the endpoint of the titration of a weak acid and weak alkali